

MATH141 Final Exam 3-Day Study Guide

Built from the uploaded scope: Tutorials 4-8, Quiz 2, final exam practice questions, and the formula booklet. This guide assumes you missed class: it starts from methods, then walks through each question type.

How to use it: first learn the method boxes, then cover the question walkthroughs with paper and redo the algebra. The final exam will likely reward method marks, so write substitutions, limits, vectors, normals, augmented matrices, and determinant expansions clearly.

Formula sheet: References are given by page/section from the uploaded MATH141 Final Exam Formula Booklet. Some page 2-3 formulas did not extract as text, but their section headings are visible: Complex numbers, 3-dimensional space, dot product, cross product, lines, planes, and distances.

Three-day crash plan

Day 1: Integration and FTC - Power/log rules, u-substitution, trig integrals, geometric area, distance from velocity, derivative of integrals with variable limits. Redo Tutorial 4, Quiz 2 Q1-Q2, Practice Q1-Q3.

Day 2: Complex numbers and vectors - Real/imaginary parts, complex square roots/quadratics, polar form, spheres, dot/cross products, projections, vector lengths. Redo Tutorial 5-6, Quiz 2 Q3-Q6, Practice Q4-Q9.

Day 3: Lines/planes and linear algebra - Parametric lines, planes, intersections, distances, Gaussian elimination, matrices, determinants, inverse existence. Redo Tutorial 7-8, Practice Q10-Q15.

Formula sheet map you should memorize

Page 1: Pythagorean identities	$\sin^2(\theta) + \cos^2(\theta) = 1$; $1 + \tan^2(\theta) = \sec^2(\theta)$; $1 + \cot^2(\theta) = \csc^2(\theta)$.
Page 1: Double-angle identities	$\sin(2\theta) = 2\sin(\theta)\cos(\theta)$; $\cos(2\theta) = \cos^2(\theta) - \sin^2(\theta) = 1 - 2\sin^2(\theta) = 2\cos^2(\theta) - 1$; $\tan(2\theta) = \frac{2\tan(\theta)}{1 - \tan^2(\theta)}$.
Page 1: Basic integration	Integral $x^n dx = x^{(n+1)/(n+1)} + C$ for $n \neq -1$; integral $1/x dx = \ln x + C$; integral $\sin x dx = -\cos x + C$; integral $\cos x dx = \sin x + C$; integral $\tan x dx = -\ln \cos x + C$.
Page 1: Fundamental Theorem of Calculus	If $F(x) = \text{integral from } a \text{ to } x \text{ of } f(t)dt$, then $F'(x) = f(x)$. With variable limits: $d/dx \text{ integral from } u(x) \text{ to } v(x) f(t)dt = v'(x)f(v(x)) - u'(x)f(u(x))$.
Page 2: Complex numbers	Use $z = a + bi$, conjugate $a - bi$, modulus $ z = \sqrt{a^2 + b^2}$, polar $z = r(\cos \theta + i \sin \theta)$.
Page 2: Dot product	$a \cdot b = a b \cos \theta$. Perpendicular means dot product 0.
Page 2: Cross product	$a \times b$ is perpendicular to a and b ; $ a \times b = a b \sin \theta$; triangle area = $1/2 PQ \times PR $.
Page 3: Lines/planes/distances	Line: $r = r_0 + t v$. Plane: $n \cdot (r - r_0) = 0$. Distance point-plane: $ Ax_0 + By_0 + Cz_0 + D / \sqrt{A^2 + B^2 + C^2}$. Distance point-line: $ (Q - P) \times v / v $.

Core concepts from zero

Integration

An antiderivative reverses differentiation. For sums, integrate term by term. Rewrite radicals as powers before using the power rule: $\sqrt{x} = x^{1/2}$, $\text{cube_root}(x^2) = x^{2/3}$. For a definite integral, find an antiderivative F and compute $F(b) - F(a)$.

Formula sheet: Page 1, Basic integration.

u-substitution

Use substitution when the integrand contains an inside function and its derivative. Set $u = \text{inside}$, replace du , change bounds for definite integrals, integrate in u , then substitute back if indefinite.

Trigonometric integrals

For powers of sin/cos: save one factor whose derivative matches the substitution. For tan/sec: save $\sec^2 x \, dx$ for $u=\tan x$, or save $\sec x \tan x \, dx$ for $u=\sec x$. For even powers of sin and cos, use double-angle identities.

Formula sheet: Page 1, Pythagorean identities and Double-angle identities.

Complex numbers

Expand using $i^2=-1$, then equate real parts and imaginary parts. For square roots, set $\sqrt{a+bi}=x+yi$, square both sides, and solve $x^2-y^2=a$ and $2xy=b$. For polar form, $r=\sqrt{a^2+b^2}$, θ is the quadrant-correct angle.

Formula sheet: Page 2, Complex numbers.

Vectors, lines, and planes

A vector from P to Q is $\mathbf{Q}-\mathbf{P}$. Dot product measures projection and angle. Cross product gives a perpendicular vector and area. A plane needs a point and normal vector. A line needs a point and direction vector. A line perpendicular to a plane uses the plane normal as its direction.

Formula sheet: Page 2, Dot product/Cross product; Page 3, Lines and Planes in 3-dimensional space.

Linear algebra

Gaussian elimination turns a system into row echelon form. Pivot variables are solved directly; free variables create infinitely many solutions. A square matrix has no inverse exactly when $\det(A)=0$. Determinants can be expanded along rows/columns with zeros.

Tutorial 4: Integration

T4 Q1a

Question: Evaluate integral $(x^2-4)/\text{cube_root}(x^2) \, dx$.

Concept: Rewrite radicals as powers and integrate term by term.

- $\text{cube_root}(x^2)=x^{(2/3)}$, so split: $x^2/x^{(2/3)}-4/x^{(2/3)}=x^{(4/3)}-4x^{(-2/3)}$.

$$\int [x^{(4/3)}-4x^{(-2/3)}] \, dx = (3/7)x^{(7/3)}-12x^{(1/3)}+C$$

Final answer: $(3/7)x^{(7/3)}-12x^{(1/3)}+C$

Formula sheet: Page 1, Basic integration power rule.

T4 Q1b

Question: Evaluate integral from -3 to 0 of $(1+\sqrt{9-x^2}) \, dx$.

Concept: Separate the easy integral and recognize a quarter circle of radius 3.

- Integral of 1 from -3 to 0 is 3.

- $\sqrt{9-x^2}$ is the upper semicircle radius 3. From -3 to 0 is one quarter of the circle.

- Quarter circle area = $(1/4)\pi(3^2)=9\pi/4$.

Final answer: $3+9\pi/4$ (about 10.069)

Formula sheet: Page 1, Basic integration for definite-integral structure; geometry is not explicitly listed on the formula sheet.

T4 Q1c

Question: Evaluate integral $x^2/\sqrt{x+1} \, dx$.

Concept: Use $u=x+1$ so the denominator becomes $\sqrt{u}$.

- Set $u=x+1$, $dx=du$, $x=u-1$.

$$\int (u-1)^2/u^{(1/2)} \, du = \int [u^{(3/2)}-2u^{(1/2)}+u^{(-1/2)}] \, du$$

- Integrate each power.

Final answer: $(2/5)(x+1)^{5/2} - (4/3)(x+1)^{3/2} + 2(x+1)^{1/2} + C$

Formula sheet: Page 1, Basic integration power rule.

T4 Q1d

Question: Evaluate integral $\sin^3(x)/\cos(x) dx$.

Concept: Save one $\sin x dx$ and use $u=\cos x$.

- Write $\sin^3 x = \sin x(1-\cos^2 x)$.

- Let $u=\cos x$, $du=-\sin x dx$.

$$\int \sin^3 x / \cos x dx = \int (u-1/u) du$$

Final answer: $(1/2)\cos^2(x) - \ln|\cos x| + C$

Formula sheet: Page 1, Pythagorean identities and Basic integration log rule.

T4 Q1e

Question: Evaluate integral $x^2 \text{cube_root}(x^3+5) dx$.

Concept: Inside function x^3+5 has derivative $3x^2$.

- Let $u=x^3+5$, $du=3x^2 dx$.

$$\int x^2 (x^3+5)^{1/3} dx = (1/3) \int u^{1/3} du$$

Final answer: $(1/4)(x^3+5)^{4/3} + C$

Formula sheet: Page 1, Basic integration power rule.

T4 Q2

Question: Work done by force $F(x)=300\sqrt{x}-10x\sqrt{x}$ from $x=0$ to $x=36$.

Concept: Work is the integral of force over distance.

- Rewrite $F(x)=300x^{1/2}-10x^{3/2}$.

- Integrate from 0 to 36.

$$W = [200x^{3/2} - 4x^{5/2}]_0^{36}$$

Final answer: 12096

Formula sheet: Page 1, Basic integration.

T4 Q3

Question: Area under $y=\sqrt{25-x^2}$ from $x=0$ to $x=5$.

Concept: This is a quarter circle of radius 5.

- $x^2+y^2=25$ is a circle radius 5.

- From 0 to 5 under the upper curve is one quarter of the full circle.

Final answer: $25\pi/4$ square meters

Formula sheet: Definite integral setup uses Page 1 FTC; circle area is geometry, not shown on formula sheet.

T4 Q4

Question: Velocity $v(t)=t-t^2$ on $0 \leq t \leq 2$. Find distance travelled.

Concept: Distance is integral of speed $|v(t)|$, not signed displacement.

- $v(t)=t(1-t)$, so it is positive on $[0,1]$ and negative on $[1,2]$.

$$\text{distance} = \int_0^1 (t-t^2) dt + \int_1^2 (t^2-t) dt$$

- The two parts are $1/6$ and $5/6$.

Final answer: 1

Formula sheet: Page 1, Basic integration/FTC.

T4 Q5a

Question: Differentiate integral from $1/x$ to $1/\sqrt{x}$ of $\sin(t^2)$ dt.

Concept: Use FTC with two variable limits.

- Let $f(t)=\sin(t^2)$, lower $u=x^{-1}$, upper $v=x^{-1/2}$.

$$\frac{d}{dx} \int_{u^v} f(t) dt = v'(x)f(v) - u'(x)f(u)$$

- $v'=-1/(2x^{3/2})$, $u'=-1/x^2$.

Final answer: $-\sin(1/x)/(2x^{3/2}) + \sin(1/x^2)/x^2$

Formula sheet: Page 1, Fundamental Theorem of Calculus.

T4 Q5b

Question: $g(x)=\int_{\cos x}^{\sin x} \ln(2+3t)$ dt. Find $g'(0)$.

Concept: Same FTC rule with trig limits.

- $g'(x)=\cos x \ln(2+3\sin x) - (-\sin x)\ln(2+3\cos x)$.

- At $x=0$: $\cos 0=1$, $\sin 0=0$.

Final answer: $\ln 2$

Formula sheet: Page 1, FTC and derivative of sin/cos.

T4 Q6a

Question: Evaluate integral $\sqrt{\sin x} \cos^5 x$ dx.

Concept: Save $\cos x$ dx; convert $\cos^4 x$ using $1-\sin^2 x$.

- Let $u=\sin x$, $du=\cos x$ dx.

- $\cos^5 x$ dx $=(\cos^4 x)(\cos x$ dx) $=(1-u^2)^2 du$.

$$\int u^{1/2} (1-2u^2+u^4) du$$

Final answer: $(2/3)\sin^{3/2}x - (4/7)\sin^{7/2}x + (2/11)\sin^{11/2}x + C$

Formula sheet: Page 1, Pythagorean identities and Basic integration.

T4 Q6b

Question: Evaluate integral $\sin^2 x \cos^2 x$ dx.

Concept: Use $\sin(2x)=2\sin x \cos x$.

- $\sin^2 x \cos^2 x = (1/4)\sin^2(2x)$.

- Using $\cos(4x)=1-2\sin^2(2x)$, $\sin^2(2x)=(1-\cos 4x)/2$.

$$\int \sin^2 x \cos^2 x dx = \int (1-\cos 4x)/8 dx$$

Final answer: $x/8 - \sin(4x)/32 + C$

Formula sheet: Page 1, Double-angle identities and Basic integration.

T4 Q6c

Question: Evaluate integral $\sin(7x)\cos(3x)$ dx.

Concept: Use product-to-sum or derive it from angle-addition identities.

$$-\sin A \cos B = \frac{1}{2}[\sin(A+B) + \sin(A-B)].$$

$$\text{integral} = \frac{1}{2} \text{integral} [\sin(10x) + \sin(4x)] dx$$

$$\text{Final answer: } -\cos(10x)/20 - \cos(4x)/8 + C$$

Formula sheet: Page 1 has trig integration basics; product-to-sum is not explicitly shown, but follows from trig identities.

Tutorial 5: Trigonometric integration and complex numbers

T5 Q1a

Question: Evaluate $\int \tan^2 x \sec^4 x \, dx$.

Concept: For sec/tan powers, save $\sec^2 x \, dx$ and set $u = \tan x$.

$$-\sec^4 x = \sec^2 x \sec^2 x = (1 + \tan^2 x) \sec^2 x.$$

$$-\text{Let } u = \tan x, du = \sec^2 x \, dx.$$

$$\text{integral } u^2(1+u^2) du$$

$$\text{Final answer: } \tan^3 x/3 + \tan^5 x/5 + C$$

Formula sheet: Page 1, Pythagorean identity $1 + \tan^2 = \sec^2$ and derivative of $\tan$.

T5 Q1b

Question: Evaluate $\int \tan^3 x \sec^3 x \, dx$.

Concept: Save $\sec x \tan x \, dx$ and set $u = \sec x$.

$$-\tan^3 \sec^3 = \tan^2 \sec^2 (\tan \sec \, dx).$$

$$-\text{Use } \tan^2 x = \sec^2 x - 1 \text{ and } u = \sec x.$$

$$\text{integral } (u^2 - 1)u^2 \, du$$

$$\text{Final answer: } \sec^5 x/5 - \sec^3 x/3 + C$$

Formula sheet: Page 1, Pythagorean identity and derivative of $\sec$ is assumed/not printed in extracted formula sheet.

T5 Q2a

Question: Find x, y : $x(1+i)^2 + y(2-i)^2 = 3+10i$.

Concept: Expand and equate real and imaginary parts.

$$-(1+i)^2 = 2i \text{ and } (2-i)^2 = 3-4i.$$

$$-\text{Equation becomes } 3y + i(2x-4y) = 3+10i.$$

$$-\text{So } 3y = 3 \text{ and } 2x-4y = 10.$$

$$\text{Final answer: } x=7, y=1$$

Formula sheet: Page 2, Complex numbers.

T5 Q2b

Question: Find x, y : $1/(x+iy) + 1/(1+2i) = 1$.

Concept: Rationalize complex fractions and equate parts.

$$-1/(1+2i) = (1-2i)/5.$$

$$-\text{So } 1/(x+iy) = 1 - (1-2i)/5 = 4/5 + 2i/5.$$

$$-\text{Invert: } x+iy = 1/(4/5 + 2i/5) = 1-i/2.$$

$$\text{Final answer: } x=1, y=-1/2$$

Formula sheet: Page 2, Complex numbers.

T5 Q2c

Question: Solve $z-8=i(7-2\text{ conjugate}(z))$; find $|z|$.

Concept: Let $z=x+iy$ and $\text{conjugate}(z)=x-iy$.

- Right side: $i(7-2x+2iy)=-2y+i(7-2x)$.

- Left side: $(x-8)+iy$.

- Equate: $x-8=-2y$ and $y=7-2x$.

Final answer: $z=2+3i$, $|z|=\sqrt{13}$

Formula sheet: Page 2, Complex numbers/modulus.

T5 Q3a

Question: Solve $2z^2-2iz-5=0$.

Concept: Use the quadratic formula with complex coefficients.

- $z=[2i \pm \sqrt{(-2i)^2-4(2)(-5)})]/4$.

- Discriminant = $-4+40=36$.

Final answer: $z=3/2+i/2$ and $z=-3/2+i/2$

Formula sheet: Page 2, Complex numbers.

T5 Q3b

Question: Solve $z^2-(1-4i)z-(5-i)=0$.

Concept: Quadratic formula, then simplify the complex square root.

- Solutions from simplification are $-1-i$ and $2-3i$.

- Check by substitution if unsure: both make the polynomial zero.

Final answer: $z=-1-i$ or $z=2-3i$

Formula sheet: Page 2, Complex numbers.

T5 Q3c

Question: Solve $z^2-z+8+2(z+1)i=0$.

Concept: Collect complex coefficients and solve.

- This is $z^2+(-1+2i)z+(8+2i)=0$.

- Quadratic simplification gives two roots.

Final answer: $z=2i$ or $z=1-4i$

Formula sheet: Page 2, Complex numbers.

T5 Q4a

Question: Express $1-i$ in polar/trigonometric form.

Concept: Find modulus and quadrant angle.

- $r=\sqrt{1^2+(-1)^2}=\sqrt{2}$.

- Point is quadrant IV, so $\theta=-\pi/4$ or $7\pi/4$.

Final answer: $\sqrt{2}[\cos(7\pi/4)+i\sin(7\pi/4)]$

Formula sheet: Page 2, Complex numbers.

T5 Q4b

Question: Express $-2+2\sqrt{3}i$ in polar form.

Concept: Find modulus and quadrant angle.

- $r = \sqrt{(-2)^2 + (2\sqrt{3})^2} = 4$.

- Quadrant II with reference angle $\pi/3$, so $\theta = 2\pi/3$.

Final answer: $4[\cos(2\pi/3) + i \sin(2\pi/3)]$

Formula sheet: Page 2, Complex numbers.

Tutorial 6: Three-dimensional space and vectors

T6 Q1a

Question: Find center/radius of $x^2 + y^2 + z^2 - 4x - 6z - 3 = 0$.

Concept: Complete the square.

- $(x^2 - 4x) = (x-2)^2 - 4$; $(z^2 - 6z) = (z-3)^2 - 9$.

- Move constants to the other side.

Final answer: Center $(2, 0, 3)$, radius 4

Formula sheet: Page 2, 3-dimensional space.

T6 Q1b

Question: Find center/radius of $x^2 + y^2 + z^2 + 2y + 4z = 20$.

Concept: Complete the square in y and z.

- $(y+1)^2 - 1$ and $(z+2)^2 - 4$.

- Equation becomes $x^2 + (y+1)^2 + (z+2)^2 = 25$.

Final answer: Center $(0, -1, -2)$, radius 5

Formula sheet: Page 2, 3-dimensional space.

T6 Q2

Question: Sphere with diameter endpoints $A(2, 1, 4)$, $B(4, 3, 10)$.

Concept: Center is midpoint; radius is half distance AB.

- Midpoint $C = ((2+4)/2, (1+3)/2, (4+10)/2) = (3, 2, 7)$.

- $AB = \sqrt{2^2 + 2^2 + 6^2} = 2\sqrt{11}$, so $r = \sqrt{11}$.

Final answer: $(x-3)^2 + (y-2)^2 + (z-7)^2 = 11$

Formula sheet: Page 2, 3-dimensional space.

T6 Q3

Question: $P(1, -1, 0)$, $Q(2, 1, -1)$, $R(-1, 1, 3)$: find PQ, PR, dot product, triangle area.

Concept: Subtract coordinates; area uses cross product.

- $PQ = \langle 1, 2, -1 \rangle$, $PR = \langle -2, 2, 3 \rangle$.

- $PQ \cdot PR = -1$.

- $PQ \times PR = \langle 8, -1, 6 \rangle$, whose length is $\sqrt{101}$.

- Triangle area is half the parallelogram area.

Final answer: Area $= \sqrt{101}/2$

T6 Q4a

Question: Given $|u|=1/2$, $|v|=2$, angle $\pi/4$, find $|u-v|$.

Concept: Expand $(u-v) \cdot (u-v)$.

$$- |u-v|^2 = |u|^2 + |v|^2 - 2|u||v|\cos(\theta).$$

$$- = (1/4) + 4 - 2(1/2)(2)\cos(\pi/4) = 17/4 - \sqrt{2}.$$

Final answer: $\sqrt{17/4 - \sqrt{2}}$

Formula sheet: Page 2, Dot product.

T6 Q4b

Question: Unit u, v with angle $2\pi/3$. Find $|u+2v|$.

Concept: Expand dot product.

$$- |u+2v|^2 = |u|^2 + 4|v|^2 + 4u \cdot v.$$

$$- u \cdot v = \cos(2\pi/3) = -1/2.$$

Final answer: $\sqrt{3}$

Formula sheet: Page 2, Dot product.

T6 Q4c

Question: Angle between $a = \langle 2, 2, -1 \rangle$ and $b = \langle 1, 2, 1 \rangle$.

Concept: Use $\cos \theta = (a \cdot b) / (|a||b|)$.

$$- a \cdot b = 5; |a| = 3; |b| = \sqrt{6}.$$

Final answer: $\theta = \arccos(5/(3\sqrt{6}))$

Formula sheet: Page 2, Dot product.

T6 Q4d

Question: Find x so $\langle 2x, x+8, 7 \rangle$ and $\langle -3, 2, x \rangle$ are perpendicular.

Concept: Perpendicular means dot product zero.

$$- \text{Dot product} = -6x + 2x + 16 + 7x = 3x + 16.$$

$$- \text{Set } 3x + 16 = 0.$$

Final answer: $x = -16/3$

Formula sheet: Page 2, Dot product.

T6 Q4e

Question: Given $\langle 5, -2 \rangle$ parallel to $\langle 15, c \rangle$, find c .

Concept: Parallel vectors are scalar multiples.

$$- 15 = 3 \cdot 5, \text{ so the second component must be } 3 \cdot (-2).$$

Final answer: $c = -6$

Formula sheet: Page 2, 3-dimensional vectors.

T6 Q5

Question: If $a \cdot b = 3$ and $a \times b = \langle -1, 0, \sqrt{8} \rangle$, find $\tan \theta$.

Concept: $\tan \theta = \sin \theta / \cos \theta = |a \times b| / (a \cdot b)$.

- $|a \times b| = \sqrt{1+8} = 3$.

- $\tan \theta = 3/3$.

Final answer: 1

Formula sheet: Page 2, Dot product and Cross product.

T6 Q6a

Question: Find vector of magnitude 7 perpendicular to $u = \langle 3, -2, 1 \rangle$ and $v = \langle 1, -2, 2 \rangle$.

Concept: Cross product gives perpendicular direction; scale to length 7.

- $u \times v = \langle -2, -5, -4 \rangle$, $\text{length} = 3\sqrt{5}$.

- Scale: $7(u \times v) / |u \times v|$.

Final answer: One answer: $7\langle -2, -5, -4 \rangle / (3\sqrt{5})$. The negative also works.

Formula sheet: Page 2, Cross product.

T6 Q6b

Question: Write $b = \langle 1, 2, 3 \rangle$ as sum of vector parallel to $a = \langle 3, 6, -2 \rangle$ and orthogonal to a .

Concept: Use projection of b onto a .

- parallel part = $\text{proj}_a b = [(b \cdot a) / (a \cdot a)]a$.

- $b \cdot a = 9$, $a \cdot a = 49$.

- Parallel part = $(9/49)\langle 3, 6, -2 \rangle = \langle 27/49, 54/49, -18/49 \rangle$.

- Orthogonal part = $b - \text{parallel} = \langle 22/49, 44/49, 165/49 \rangle$.

Final answer: $b = \langle 27/49, 54/49, -18/49 \rangle + \langle 22/49, 44/49, 165/49 \rangle$

Formula sheet: Page 2, Dot product/projection.

Tutorial 7: Lines and planes in space

T7 Q1a

Question: Plane through $P(4, 3, 6)$, $Q(-2, 0, 8)$, $R(1, 5, 0)$.

Concept: Two direction vectors in the plane; cross product is normal.

- $PQ = \langle -6, -3, 2 \rangle$, $PR = \langle -3, 2, -6 \rangle$.

- $PQ \times PR = \langle 14, -42, -21 \rangle$, simplify to $\langle 2, -6, -3 \rangle$.

- Use $n \cdot (x - P) = 0$.

Final answer: $2x - 6y - 3z + 28 = 0$

Formula sheet: Page 2 Cross product; Page 3 Planes.

T7 Q1b

Question: Plane through $(1, 2, -3)$ perpendicular to line $x = 1 + 2t, y = 2 + t, z = -3 - 5t$.

Concept: A plane perpendicular to a line has normal equal to the line direction.

- Line direction is $\langle 2, 1, -5 \rangle$.

- Use $\langle 2, 1, -5 \rangle \cdot \langle x - 1, y - 2, z + 3 \rangle = 0$.

Final answer: $2x + y - 5z - 19 = 0$

T7 Q2

Question: Line through $P(1, -1, 5)$ and $Q(0, 1, 3)$.

Concept: Direction vector is $Q-P$.

- $Q-P = \langle -1, 2, -2 \rangle$.

Final answer: $x=1-t, y=-1+2t, z=5-2t$

T7 Q3

Question: Line of intersection of planes $4x+4y-2z=9$ and $2x+y+z=-3$.

Concept: Direction is cross product of plane normals; find one point satisfying both planes.

- Normals $n_1 = \langle 4, 4, -2 \rangle$, $n_2 = \langle 2, 1, 1 \rangle$. Cross product can use $\langle 6, -8, -4 \rangle$.

- Set $z=0$, solve $4x+4y=9$ and $2x+y=-3$ to get $x=-21/4, y=15/2$.

Final answer: $x=-21/4+6t, y=15/2-8t, z=-4t$

T7 Q4

Question: Intersection of line $x=1-3t, y=1-t, z=-1+2t$ with plane $-x+3y+3z=1$.

Concept: Substitute line into plane, solve for t .

- $-(1-3t)+3(1-t)+3(-1+2t)=1$.

- This gives $3t=1$, so $t=1/3$.

Final answer: $(0, 2/3, -1/3)$

T7 Q5a

Question: Line through $(2, 4, 1)$ perpendicular to plane $3x-y+5z=77$.

Concept: Plane normal is the direction of the perpendicular line.

- Normal vector is $\langle 3, -1, 5 \rangle$.

Final answer: $x=2+3t, y=4-t, z=1+5t$

T7 Q5b

Question: Angle between planes $x-2y+z=5$ and $2x+y-z=0$.

Concept: Angle between planes equals angle between normals.

- $n_1 = \langle 1, -2, 1 \rangle$, $n_2 = \langle 2, 1, -1 \rangle$.

- $|n_1 \cdot n_2| = 1$; $|n_1| = |n_2| = \sqrt{6}$.

- $\cos \theta = 1/6$.

Final answer: $\theta = \arccos(1/6)$

T7 Q6a

Question: Distance from A(1,-1,3) to plane $-2x+y-2z=1$.

Concept: Use point-plane distance formula.

- Write plane as $-2x+y-2z-1=0$.

- Distance = $|-2(1)+(-1)-2(3)-1|/\sqrt{4+1+4}$.

Final answer: $10/3$

Formula sheet: Page 3, Distances in 3-dimensional space.

T7 Q6b

Question: Distance between planes $z=x+2y+1$ and $3x+6y-3z=4$.

Concept: Rewrite with same normal and compare constants.

- First: $x+2y-z+1=0$.

- Second: $x+2y-z-4/3=0$.

- Distance = $|1-(-4/3)|/\sqrt{1+4+1}$.

Final answer: $7/(3\sqrt{6})$

Formula sheet: Page 3, Distances in 3-dimensional space.

T7 Q6c

Question: Distance from Q(3,-1,4) to line $x=-2+3t, y=-2t, z=1+4t$.

Concept: Use point-line distance formula.

- Point on line $P=(-2,0,1)$, direction $v=\langle 3,-2,4 \rangle$.

- Distance = $|(Q-P) \times v|/|v|$.

Final answer: $\sqrt{6}$

Formula sheet: Page 3, Distances in 3-dimensional space.

Tutorial 8: Linear algebra

T8 Q1

Question: Solve system $x_1+x_2-2x_3=9$; $2x_1+4x_2-3x_3=1$; $3x_1+6x_2-5x_3=0$.

Concept: Use Gaussian elimination on augmented matrix.

- Row reduction gives pivots in x_1, x_2, x_3 .

- The reduced solution is $x_1=25, x_2=-10, x_3=3$.

Final answer: $(25, -10, 3)$

Formula sheet: Linear algebra method; determinant/system formulas are not visible in extracted formula sheet.

T8 Q2

Question: Show a 3-equation, 4-variable system has infinitely many solutions and find set.

Concept: A free variable gives infinitely many solutions.

- RREF is: $x_1+2x_4=3$; $x_2-3x_4=-5$; $x_3+4x_4=7$.

- Let $x_4=t$.

- Then $x_1=3-2t, x_2=-5+3t, x_3=7-4t$.

Final answer: $(x_1, x_2, x_3, x_4) = (3-2t, -5+3t, 7-4t, t)$

Formula sheet: Linear algebra row reduction.

T8 Q3a1

Question: For A 3×2 and B 2×3 , compute $A+B^T$.

Concept: Dimensions match after transpose: B^T is 3×2 .

- Add corresponding entries.

Final answer: $\begin{bmatrix} 0 & 4 \\ 3 & -1 \\ 8 & -8 \end{bmatrix}$

Formula sheet: Matrix arithmetic.

T8 Q3a2

Question: Compute AC.

Concept: A is 3×2 and C is 2×2 , so product is 3×2 .

- Multiply rows of A by columns of C.

Final answer: $\begin{bmatrix} 2 & 0 \\ 7 & 1 \\ 21 & 3 \end{bmatrix}$

Formula sheet: Matrix multiplication.

T8 Q3b

Question: Find X such that $(\frac{3}{2})X+C=\begin{bmatrix} 3 & -4 \\ 5 & 4 \end{bmatrix}$.

Concept: Isolate X like algebra, but matrix entries move together.

- $(\frac{3}{2})X = \text{target}-C = \begin{bmatrix} 1 & -4 \\ 8 & 5 \end{bmatrix}$.

- Multiply by $\frac{2}{3}$.

Final answer: $X=\begin{bmatrix} 2/3 & -8/3 \\ 16/3 & 10/3 \end{bmatrix}$

Formula sheet: Matrix arithmetic.

T8 Q4

Question: Find h so A^{-1} does not exist for the given 4×4 matrix.

Concept: Inverse fails when determinant is zero.

- $\det(A)=12(h+6)$.

- Set $12(h+6)=0$.

Final answer: $h=-6$

Formula sheet: Determinants/inverse criterion.

T8 Q5

Question: Use cofactor expansion to find determinant of the given 4×4 matrix.

Concept: Expand along the second column because it has three zeros.

- Only entry in column 2 is 1 at row 2, so $\det(A)=(-1)^{(2+2)}\det(\text{minor})$.

- The 3×3 minor determinant is 36.

Final answer: 36

Formula sheet: Cofactor expansion/determinants.

T8 Q6

Question: Given $\det(A)=9$ and $\det(B)=4$, evaluate determinant expressions.

Concept: Use determinant properties.

- $\det(-3A) = (-3)^3 \det(A) = -27 \cdot 9 = -243$ for 3×3 A .

- $\det(BA^{-1}) = \det(B)\det(A^{-1}) = 4 \cdot (1/9) = 4/9$.

- $\det$ with rows $[\text{row}_2; \text{row}_3; 2\text{row}_1]$ is a cyclic row permutation of A with row_1 scaled by 2, so $\det = 2\det(A) = 18$.

- For rows $[-2\text{row}_1; \text{row}_2; \text{row}_3 - \text{row}_2]$, adding/subtracting a multiple of one row to another does not change determinant after scaling row_1 by -2, so $\det = -2\det(A) = -18$.

Final answer: 1) -243; 2) $4/9$; 3) 18; 4) -18

Formula sheet: Determinant properties.

Quiz 2: exam-style walkthrough

Quiz Q1a

Question: Evaluate integral $(x^4 + x^2 - 2)/x^3 \, dx$.

Concept: Divide each term by x^3 first.

- Integrand becomes $x + 1/x - 2x^{-3}$.

- Integrate term by term.

Final answer: $x^2/2 + \ln|x| + 1/x^2 + C$

Formula sheet: Page 1, Basic integration.

Quiz Q1b(i)

Question: Evaluate integral from 0 to 3 of $x \sqrt{x^2 + 1} \, dx$.

Concept: Use $u = x^2 + 1$.

- $du = 2x \, dx$; bounds: $x=0 \rightarrow u=1$, $x=3 \rightarrow u=10$.

- $\text{Integral} = (1/2) \int_1^{10} u^{1/2} du$.

Final answer: $(1/3)(10^{3/2} - 1)$

Formula sheet: Page 1, Basic integration/u-substitution.

Quiz Q1b(ii)

Question: Evaluate integral $\sin^3 x \cos^3 x \, dx$.

Concept: Use $\sin^2 = 1 - \cos^2$ and save $\sin x \, dx$.

- $\sin^3 \cos^3 = \sin^2 \times \sin x \cos^3 x$.

- Let $u = \cos x$, $du = -\sin x \, dx$.

- Integral becomes $\int (u^5 - u^3) du$.

Final answer: $\cos^6 x / 6 - \cos^4 x / 4 + C$

Formula sheet: Page 1, Pythagorean identities and Basic integration.

Quiz Q2

Question: $F(x) = \int_2^{3x^2} \sqrt{t^2 + 25} \, dt$. Find $F'(2)$.

Concept: FTC with upper limit only.

- Upper limit $v = 3x^2$, $v' = 6x$.

- $F'(x) = 6x \sqrt{(3x^2)^2 + 25} = 6x \sqrt{9x^4 + 25}$.

- At $x=2$: $12\sqrt{169}$.

Final answer: 156

Formula sheet: Page 1, Fundamental Theorem of Calculus.

Quiz Q3a

Question: Find real x, y if $(2+i)x + (1-2i)y = 8-i$.

Concept: Equate real and imaginary parts.

- Real: $2x+y=8$. Imaginary: $x-2y=-1$.

- Solve the two equations.

Final answer: $x=3, y=2$

Formula sheet: Page 2, Complex numbers.

Quiz Q3b

Question: Find $\sqrt{7+24i}$.

Concept: Let $\sqrt{7+24i}=a+bi$ and square.

- $a^2-b^2=7$ and $2ab=24$, so $ab=12$.

- $a=4, b=3$ works.

Final answer: $4+3i$

Formula sheet: Page 2, Complex numbers.

Quiz Q4

Question: Sphere with diameter endpoints $A(-4,-1,2)$, $B(2,3,6)$.

Concept: Midpoint is center; half distance is radius.

- Center $=((-4+2)/2, (-1+3)/2, (2+6)/2) = (-1, 1, 4)$.

- $AB = \sqrt{6^2 + 4^2 + 4^2} = \sqrt{68}$, so $r^2=17$.

Final answer: $(x+1)^2 + (y-1)^2 + (z-4)^2 = 17$

Formula sheet: Page 2, 3-dimensional space.

Quiz Q5

Question: Find a unit vector perpendicular to $u = \langle 1, 1, 1 \rangle$ and $v = \langle 1, -1, 0 \rangle$.

Concept: Cross product gives perpendicular direction, then normalize.

- $u \times v = \langle 1, 1, -2 \rangle$.

- Length $= \sqrt{6}$.

- Unit vector $= (u \times v) / \sqrt{6}$.

Final answer: $\langle 1/\sqrt{6}, 1/\sqrt{6}, -2/\sqrt{6} \rangle$ (or its negative)

Formula sheet: Page 2, Cross product.

Quiz Q6

Question: Given $|a|=3$, $|b|=2$, $\theta=2\pi/3$, find $|a+b|$.

Concept: Expand dot product.

- $|a+b|^2 = |a|^2 + 2|a||b|\cos\theta + |b|^2$.

- $= 9 + 12(-1/2) + 4 = 7$.

Final answer: $\sqrt{7}$

Final exam practice questions walkthrough

Practice Q1.1

Question: Integrate $(x \sqrt{x} + 3x - 1) dx$.

Concept: Rewrite $x \sqrt{x} = x^{3/2}$, integrate powers.

$$\int [x^{3/2} + 3x - 1] dx$$

Final answer: $(2/5)x^{5/2} + (3/2)x^2 - x + C$

Formula sheet: Page 1, Basic integration.

Practice Q1.2

Question: Evaluate $\int_1^2 (2/x - 3/x^2) dx$.

Concept: Use log rule and power rule.

- Integral is $[2\ln|x| + 3/x]_1^2$.

Final answer: $2\ln 2 - 3/2$

Formula sheet: Page 1, Basic integration.

Practice Q2.1

Question: Evaluate $\int_0^1 x^3 \sqrt{x^4 + 1} dx$.

Concept: $u = x^4 + 1$; $du = 4x^3 dx$.

- Bounds $0 \rightarrow 1$, $1 \rightarrow 2$.

- $\text{Integral} = (1/4) \int_1^2 u^{1/2} du$.

Final answer: $(1/6)(2^{3/2} - 1)$

Formula sheet: Page 1, Basic integration/u-substitution.

Practice Q2.2

Question: Evaluate $\int \sin^3 x \sqrt{\cos x} dx$.

Concept: Save $\sin x dx$, use $u = \cos x$.

- $\sin^3 = \sin^2 \sin = (1 - \cos^2) \sin$.

- $u = \cos x$, $du = -\sin x dx$.

- Integral becomes $\int (u^2 - 1)u^{1/2} du$.

Final answer: $(2/7)\cos^{7/2} x - (2/3)\cos^{3/2} x + C$

Formula sheet: Page 1, Pythagorean identity and Basic integration.

Practice Q3

Question: $F(x) = \int_{x^2}^{5x} 1/\sqrt{t+1} dt$, find $F'(1)$.

Concept: FTC with both limits.

- $F'(x) = 5/\sqrt{5x+1} - 2x/\sqrt{x^2+1}$.

- At $x=1$.

Final answer: $5/\sqrt{6} - \sqrt{2}$

Formula sheet: Page 1, Fundamental Theorem of Calculus.

Practice Q4.1

Question: Find x, y if $(x+iy)(2+i)=3-i$.

Concept: Expand and equate parts.

- $(x+iy)(2+i)=(2x-y)+i(x+2y)$.

- $2x-y=3, x+2y=-1$.

Final answer: $x=1, y=-1$

Formula sheet: Page 2, Complex numbers.

Practice Q4.2

Question: Find x, y if $x+2iy-8=i(7-2x+2iy)$.

Concept: Equate real and imaginary parts.

- Right side $= -2y+i(7-2x)$.

- $x-8=-2y$ and $2y=7-2x$.

Final answer: $x=-1, y=9/2$

Formula sheet: Page 2, Complex numbers.

Practice Q5

Question: Find $\sqrt{-5-12i}$.

Concept: Let $a+bi$, square, solve.

- $a^2-b^2=-5$ and $2ab=-12$.

- Pairs are $(-2,3)$ or $(2,-3)$; principal square root convention uses real part sign suitable to the solution shown.

Final answer: $-2+3i$

Formula sheet: Page 2, Complex numbers.

Practice Q6

Question: Show $x^2+y^2+z^2=4x-6y+2z+1$ is a sphere and find radius.

Concept: Complete squares.

- Move all terms left: $x^2-4x+y^2+6y+z^2-2z=1$.

- Complete: $(x-2)^2+(y+3)^2+(z-1)^2=15$.

Final answer: Center $(2,-3,1)$, radius $\sqrt{15}$

Formula sheet: Page 2, 3-dimensional space.

Practice Q7

Question: Sphere with diameter endpoints $A(-1,2,3)$, $B(2,0,5)$.

Concept: Midpoint and half distance.

- Center $= (1/2, 1, 4)$.

- $AB=\sqrt{17}$, so $r=\sqrt{17}/2$.

Final answer: $(x-1/2)^2+(y-1)^2+(z-4)^2=17/4$

Formula sheet: Page 2, 3-dimensional space.

Practice Q8

Question: Given $|a|=2$, $|b|=\sqrt{2}$, $\theta=\pi/4$, find $|a+b|$.

Concept: Use $|a+b|^2$ formula.

$$= 4 + 2 + 2(2)(\sqrt{2})\cos(\pi/4) = 10.$$

Final answer: $\sqrt{10}$

Formula sheet: Page 2, Dot product.

Practice Q9.1

Question: Find x so $\langle 2x-1, 3x, 1/2 \rangle$ and $\langle 3, -2, 2x \rangle$ are perpendicular.

Concept: Dot product zero.

$$- 3(2x-1) - 2(3x) + (1/2)(2x) = 0.$$

$$- x - 3 = 0.$$

Final answer: $x = 3$

Formula sheet: Page 2, Dot product.

Practice Q9.2

Question: Area of triangle $P(1, 3, 5)$, $Q(-2, 1, 2)$, $R(1, 1, 1)$.

Concept: $\text{Area} = 1/2 |PQ \times PR|$.

$$- PQ = \langle -3, -2, -3 \rangle, PR = \langle 0, -2, -4 \rangle.$$

$$- PQ \times PR = \langle 2, -12, 6 \rangle.$$

$$- \text{Half the length is } \sqrt{46}.$$

Final answer: $\sqrt{46}$

Formula sheet: Page 2, Cross product.

Practice Q10

Question: Line of intersection of planes $4x + 4y - 2z = 9$ and $2x + y + z = -3$.

Concept: Same method as Tutorial 7 Q3.

$$- \text{Direction from cross product of normals is } \langle 6, -8, -4 \rangle.$$

$$- \text{Point with } z=0 \text{ is } (-21/4, 15/2, 0).$$

Final answer: $x = -21/4 + 6t$, $y = 15/2 - 8t$, $z = -4t$

Formula sheet: Page 3, Lines/Planes; Page 2 Cross product.

Practice Q11a

Question: Plane containing $A(2, 0, -1)$, $B(1, -1, 1)$, $C(0, 3, -2)$.

Concept: Normal is $AB \times AC$.

$$- AB = \langle -1, -1, 2 \rangle, AC = \langle -2, 3, -1 \rangle.$$

$$- AB \times AC = \langle -5, -5, -5 \rangle, \text{ equivalent normal } \langle 1, 1, 1 \rangle.$$

Final answer: $x + y + z = 1$

Formula sheet: Page 2 Cross product; Page 3 Planes.

Practice Q11b

Question: Distance from $D(5, -2, -1)$ to plane $x + y + z = 1$.

Concept: Use point-plane distance.

$$- \text{Plane: } x + y + z - 1 = 0.$$

- Distance= $|5-2-1-1|/\sqrt{3}$.

Final answer: $1/\sqrt{3}$

Formula sheet: Page 3, Distances.

Practice Q12

Question: Solve system $3x-y-5z=3$; $4x-4y-3z=-4$; $x-5z=2$.

Concept: Gaussian elimination.

- The reduced echelon form gives pivots $x=2, y=3, z=0$.

Final answer: $x=2, y=3, z=0$

Formula sheet: Linear algebra row reduction.

Practice Q13

Question: Truss system: $2x_1-2x_2+x_3=3$; $3x_1+x_2-x_3=7$; $x_1-3x_2+2x_3=0$.

Concept: Gaussian elimination.

- Row reduction gives $x_1=2, x_2=0, x_3=-1$.

Final answer: $x_1=2, x_2=0, x_3=-1$

Formula sheet: Linear algebra row reduction.

Practice Q14

Question: Find determinant of 4×4 matrix with second row $[0,0,3,0]$.

Concept: Expand along row with zeros.

- Only the 3 contributes; include sign $(-1)^{(2+3)}=-1$.

- The provided expansion simplifies to -6.

Final answer: -6

Formula sheet: Determinants/cofactor expansion.

Practice Q15a1

Question: Compute $A+B^T$ for given matrices.

Concept: Transpose B then add.

- B^T has shape 3×2 , same as A.

Final answer: $[[0,4],[3,-1],[8,-8]]$

Formula sheet: Matrix arithmetic.

Practice Q15a2

Question: Compute AC.

Concept: Matrix multiplication row by column.

- A is 3×2 and C is 2×2 , so AC is 3×2 .

Final answer: $[[2,0],[7,1],[21,3]]$

Formula sheet: Matrix multiplication.

Practice Q15b

Question: Solve $(3/2)X+C=[[3,-4],[5,4]]$.

Concept: Subtract C and multiply by $\frac{2}{3}$.

- Target-C= $\begin{bmatrix} 1, -4 \\ 8, 5 \end{bmatrix}$.

Final answer: X= $\begin{bmatrix} \frac{2}{3}, -\frac{8}{3} \\ \frac{16}{3}, \frac{10}{3} \end{bmatrix}$

Formula sheet: Matrix arithmetic.

Final checklist before the exam

- When integrating, rewrite first, then choose power rule or substitution.
- For definite u-substitution, change the bounds immediately.
- For distance travelled, integrate absolute velocity by splitting where $v(t)=0$.
- For FTC with variable limits, upper contribution is positive, lower contribution is negative.
- For complex equations, always separate real and imaginary parts.
- For a sphere, complete the square until it looks like $(x-a)^2+(y-b)^2+(z-c)^2=r^2$.
- For planes, find a normal vector; for lines, find a direction vector.
- For triangle area and perpendicular vectors, use cross product.
- For perpendicular vectors, dot product equals zero.
- For inverse existence, determinant must be nonzero; no inverse means determinant equals zero.
- In row reduction, name free variables clearly if there are more variables than pivots.